

# Hyperparameter Optimisation: A Comparative Study of Five Algorithms on YAHPO Gym

## Hyperparameter Optimisation for Machine Learning — Assignment 1

Shivam Sen Gupta (473710) and Dlyaver Djebarov (423901)

RWTH Aachen University, Germany

shivam.sengupta@rwth-aachen.de, dlyaver.djebarov@rwth-aachen.de

### 1 Introduction

We compare Random Search (RS), Grid Search (GS), Successive Halving (SH), Hyperband (HB), and Bayesian Optimisation (BO) on two YAHPO Gym surrogate benchmarks [4]: NB301 (neural architecture search, CIFAR-10) and rbv2\_xgboost (XGBoost tuning, dataset 16). HB achieves the best final accuracy on NB301; on rbv2\_xgboost the multi-fidelity methods converge faster without improving final quality.

### 2 Methods

*Random Search (RS).* RS [1] samples configurations uniformly at random and evaluates each at maximum fidelity, ignoring all prior observations.

*Grid Search (GS).* GS discretises each continuous dimension into 5 evenly-spaced values crossed with all categorical choices; NB301's 34-dimensional categorical space is handled by sampling each dimension independently.

*Successive Halving (SH).* SH [2] starts with  $n = \lceil \eta^{s_{\max}+1} / (s_{\max} + 1) \rceil$  configurations at  $r_{\min}$ , keeps the top  $1/\eta$  fraction per round, and promotes survivors to  $r \cdot \eta$  until the highest rung below  $r_{\max}$  ( $\eta=3$ ,  $s_{\max} = \lfloor \log_{\eta}(r_{\max}/r_{\min}) \rfloor$ ).

*Hyperband (HB).* HB [3] runs  $s_{\max}+1$  SH brackets with  $(n_s = \lceil (B/r_{\max}) \cdot \eta^s / (s+1) \rceil)$ ,  $r_s = \max(r_{\min}, r_{\max} \cdot \eta^{-s})$ , hedging against the sensitivity of any single bracket to its initial candidate count.

*Bayesian Optimisation (BO).* BO [5] fits a GP (Matérn-5/2 + WhiteKernel) to all past evaluations and selects the next configuration by maximising Expected Improvement over 1 000 random candidates (5 random warm-up evaluations; GP and EI implemented from scratch in NumPy).

### 3 Experimental Setup

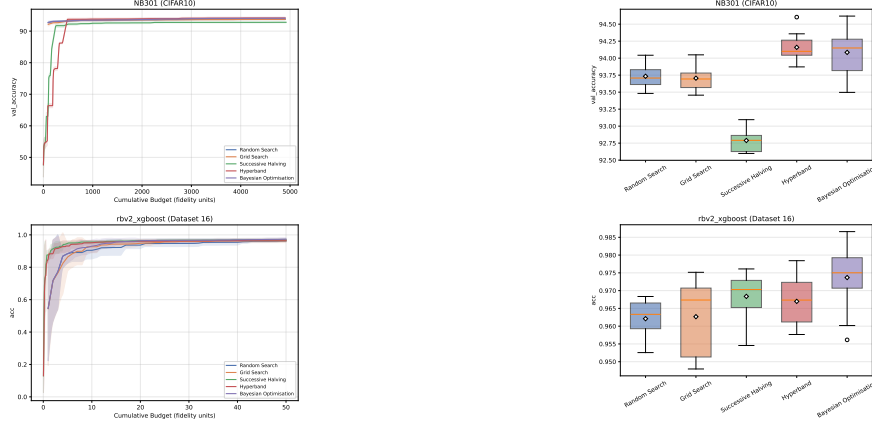
**NB301:** fidelity = epochs ( $r_{\min}=1$ ,  $r_{\max}=98$ ), metric = val. accuracy. **rbv2\_xgboost/16:** fidelity = training fraction, metric = accuracy. Budget  $B=50 \times r_{\max}$  fidelity units for all algorithms. Hyperparameter spaces accessed via ConfigSpace without the forbidden sampling methods; 10 independent random seeds per algorithm.

### 4 Results

**NB301.** All methods reach  $\approx 93\%$  within 500 budget units (Figure 1) but diverge afterwards. HB achieves the highest final accuracy ( $94.16 \pm 0.20$ ); SH is weakest ( $92.79 \pm 0.17$ ) because a single bracket exhausts its candidate pool and restarts with no memory, whereas HB always has other brackets running at different fidelity levels. BO matches HB in mean ( $94.08$ ) but is less consistent ( $\text{std} = 0.32$ , Figure 2); NB301's 34 categorical hyperparameters produce a  $\approx 272$ -dimensional one-hot vector that degrades the GP, making results highly seed-dependent. RS and GS perform nearly identically, as expected.

**Table 1:** Final incumbent (mean  $\pm$  std, 10 seeds). Best in **bold**.

| Algorithm     | NB301                              | rbv2_xgb                            |
|---------------|------------------------------------|-------------------------------------|
| Random Search | 93.73 $\pm$ 0.17                   | 0.962 $\pm$ 0.005                   |
| Grid Search   | 93.71 $\pm$ 0.18                   | 0.963 $\pm$ 0.010                   |
| Succ. Halving | 92.79 $\pm$ 0.17                   | 0.968 $\pm$ 0.007                   |
| Hyperband     | <b>94.16 <math>\pm</math> 0.20</b> | 0.967 $\pm$ 0.007                   |
| Bayesian Opt. | 94.08 $\pm$ 0.32                   | <b>0.974 <math>\pm</math> 0.009</b> |

**Figure 1:** Incumbent curves (mean  $\pm$  1 std). Top: NB301. **Figure 2:** Final incumbent per seed. HB best on NB301; BO widest spread.

*rbv2\_xgboost*. BO achieves the highest final accuracy ( $0.974 \pm 0.009$ ), but the more interesting story is in convergence speed (Figure 1). SH and HB reach  $\approx 0.96$  after only 10 budget units, while RS, GS, and BO need 20–30 units to match that level. XGBoost rankings at small training fractions correlate well with full-data rankings, so multi-fidelity methods can cheaply discard poor configurations early and reach a strong incumbent fast.

## 5 Conclusions

HB achieves the best final accuracy on NB301; its bracket portfolio hedges against a poor initial candidate count, the key advantage over a single SH bracket. On *rbv2\_xgboost*, BO reaches the highest final accuracy ( $0.974 \pm 0.009$ ), but SH and HB find a good incumbent significantly faster, making them the practical choice when convergence speed matters. BO is unreliable on NB301 due to the  $\approx 272$ -dimensional one-hot encoding degrading the GP surrogate, leading to high variance across seeds.

## Statement of Contributions

Shivam Sen Gupta implemented Random Search, Grid Search, and Bayesian Optimisation, ran all experiments, and wrote the report. Dlyaver Djebarov implemented Successive Halving and Hyperband, and contributed to the presentation slides and result visualisations.

## Bibliography

- [1] J. Bergstra and Y. Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13:281–305, 2012.
- [2] K. Jamieson and A. Talwalkar. Non-stochastic best arm identification and hyperparameter optimization. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2016.
- [3] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar. Hyperband: A novel bandit-based approach to hyperparameter optimization. *Journal of Machine Learning Research*, 18(185): 1–52, 2018.
- [4] F. Pfisterer, L. Schneider, J. Moosbauer, M. Binder, and B. Bischl. YAHPO Gym — an efficient multi-objective multi-fidelity benchmark for hyperparameter optimization. In *International Conference on Automated Machine Learning (AutoML)*, 2022.
- [5] J. Snoek, H. Larochelle, and R. P. Adams. Practical bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.

## Eidesstattliche Versicherung

### Statutory Declaration in Lieu of an Oath

\_\_\_\_\_  
Name, Vorname/Last Name, First Name

\_\_\_\_\_  
Matrikelnummer (freiwillige Angabe)

Matriculation No. (optional)

Ich versichere hiermit an Eides Statt, dass ich die vorliegende Arbeit/Bachelorarbeit/  
Masterarbeit\* mit dem Titel

I hereby declare in lieu of an oath that I have completed the present paper/Bachelor thesis/Master thesis\* entitled

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

selbstständig und ohne unzulässige fremde Hilfe (insbes. akademisches Ghostwriting)  
erbracht habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt.  
Für den Fall, dass die Arbeit zusätzlich auf einem Datenträger eingereicht wird, erkläre ich,  
dass die schriftliche und die elektronische Form vollständig übereinstimmen. Die Arbeit hat in  
gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

independently and without illegitimate assistance from third parties (such as academic ghostwriters). I have used no other than  
the specified sources and aids. In case that the thesis is additionally submitted in an electronic format, I declare that the written  
and electronic versions are fully identical. The thesis has not been submitted to any examination body in this, or similar, form.

\_\_\_\_\_  
Ort, Datum/City, Date

\_\_\_\_\_  
Unterschrift/Signature

\*Nichtzutreffendes bitte streichen

\*Please delete as appropriate

#### Belehrung:

##### Official Notification:

#### § 156 StGB: Falsche Versicherung an Eides Statt

Wer vor einer zur Abnahme einer Versicherung an Eides Statt zuständigen Behörde eine solche Versicherung  
falsch abgibt oder unter Berufung auf eine solche Versicherung falsch aussagt, wird mit Freiheitsstrafe bis zu drei  
Jahren oder mit Geldstrafe bestraft.

#### Para. 156 StGB (German Criminal Code): False Statutory Declarations

Whoever before a public authority competent to administer statutory declarations falsely makes such a declaration or falsely  
testifies while referring to such a declaration shall be liable to imprisonment not exceeding three years or a fine.

#### § 161 StGB: Fahrlässiger Falscheid; fahrlässige falsche Versicherung an Eides Statt

(1) Wenn eine der in den §§ 154 bis 156 bezeichneten Handlungen aus Fahrlässigkeit begangen worden ist, so  
tritt Freiheitsstrafe bis zu einem Jahr oder Geldstrafe ein.

(2) Strafflosigkeit tritt ein, wenn der Täter die falsche Angabe rechtzeitig berichtet. Die Vorschriften des § 158  
Abs. 2 und 3 gelten entsprechend.

#### Para. 161 StGB (German Criminal Code): False Statutory Declarations Due to Negligence

(1) If a person commits one of the offences listed in sections 154 through 156 negligently the penalty shall be imprisonment not  
exceeding one year or a fine.

(2) The offender shall be exempt from liability if he or she corrects their false testimony in time. The provisions of section 158 (2)  
and (3) shall apply accordingly.

Die vorstehende Belehrung habe ich zur Kenntnis genommen:

I have read and understood the above official notification:

\_\_\_\_\_  
Ort, Datum/City, Date

\_\_\_\_\_  
Unterschrift/Signature